

Rhenium diselenide (ReSe₂) infrared photodetector enhanced by (3-aminopropyl)trimethoxysilane (APTMS) treatment

Muhammad Hasnain Ali^{a,1}, Dong-Ho Kang^{a,1}, Jin-Hong Park^{a,b,*}

^a School of Electronic and Electrical Engineering, Sungkyunkwan University, Suwon 440-746, South Korea

^b SKKU Advanced Institute of Nano Technology (SAINT), Suwon 440-746, South Korea

ARTICLE INFO

Keywords:

TMD
ReSe₂
Infrared photodetector
Organic molecule doping
APTMS

ABSTRACT

Rhenium diselenide (ReSe₂) has a small bandgap (< 1 eV), which facilitates its longer wavelength photo-detection capability. However, ReSe₂-based photodetectors have presented relatively low photoresponsivity compared to devices fabricated on conventional transition metal dichalcogenides (TMDs), such as MoS₂ and WSe₂. Here, we demonstrate a high-performance ReSe₂ near-infrared photodetector with high photoresponsivity (4.2×10^4 A/W), a wide detection range (520–1064 nm), and fast photoswitching time (τ_{rise} : 1.9 ms and τ_{decay} : 4.5 ms). This was accomplished by an organic molecule-based n-doping technique based on a (3-aminopropyl) trimethoxysilane (APTMS). The n-doping concentration of ReSe₂ was controlled between 7.82×10^{10} and 3.90×10^{11} cm⁻², which lies in the nondegenerate regime. After the APTMS treatment, photoresponsivity values were increased from 4.3×10^3 to 4.2×10^4 A/W (tenfold at $\lambda = 785$ nm) and from 3.6 to 28.5 A/W (eightfold at $\lambda = 1064$ nm). The photoswitching times obtained after the n-type doping ($\tau_{\text{rise}}/\tau_{\text{decay}}$: 1.9/4.5 ms) were also shorter than previously reported values in the devices formed on rhenium-based TMDs. We expect that this doping technique would be useful for the integration and performance optimization of various TMD-based devices in the future.

1. Introduction

Transition metal dichalcogenides (TMDs) have recently emerged as promising two-dimensional (2D) materials for transparent, flexible, and wearable electronics, because of their novel electrical, optical, and mechanical properties [1–3]. In particular, the van der Waals epitaxial structure of TMDs without dangling bonds allows TMD-based transistors to have outstanding scalability of channel thickness (approximately 0.7 nm) [4]. In addition, owing to their exceptional quantum efficiency and high photoluminescence, high-performance photodetectors have been demonstrated on TMDs [5–9]. However, conventional TMDs (Group VI MX₂: M = molybdenum or tungsten, X = sulfur or selenium) have exhibited poor detection capabilities in the infrared region (> 800 nm wavelength) because their bandgap is larger than 1.2 eV [10,11]. Although, black phosphorous (E_g = about 0.3 eV) and graphene (theoretically, no E_g) have shown detection in the infrared region, the achieved photoresponsivity values are significantly low owing to limited optical absorption and short lifetime of photogenerated carriers (5 mA/W and 0.1–6 mA/W at 1550 nm, respectively) [12–14]. In this light, rhenium diselenide (ReSe₂) with a relatively narrow bandgap (0.9–1.0 eV) [15] is a prospective candidate for realizing a 2D

infrared photodetector, but this shows low photoresponsivity relative to other TMD-based devices. Yang et al. reported monolayer ReSe₂ devices with photoresponsivity of 55.5 AW⁻¹ (by molybdenum-based substitution doping during synthesis) [16] and 95 AW⁻¹ (by O₂ molecular physisorption-based gating in an O₂ environment) [15]. These values are considerably lower than 880 AW⁻¹ and 2200 AW⁻¹ of MoS₂ photodetectors demonstrated by Lopez-Sanchez et al. [17] and Zhang et al. [18].

Recently, polymer- and chemical-based doping of TMDs has been proposed as a method for improving the performance of TMD-based photodetectors [19–23]. In addition, considerable enhancement in photoresponsivity was also achieved via the plasma doping of TMD [24]. Here, we demonstrate a ReSe₂-based photodetector with high photoresponsivity and longer wavelength detection capability, which was achieved by applying an organic molecule doping technique based on (3-aminopropyl)trimethoxysilane (APTMS). The negative dipoles of the amine groups (–NH₂) in APTMS are expected to induce n-doping of ReSe₂. The doping effects on ReSe₂-based devices are investigated by Raman spectroscopy and I–V measurements (under dark and light illumination).

* Corresponding author. School of Electronic and Electrical Engineering, Sungkyunkwan University, Suwon 440-746, South Korea.

E-mail address: jhpark9@skku.edu (J.-H. Park).

¹ These authors are equally contributed in this work.

2. Experiments

2.1. Fabrication of ReSe₂ devices

First, ReSe₂ multilayer flakes were mechanically exfoliated on adhesive Scotch tape from bulk ReSe₂ crystals and transferred onto a heavily p-doped silicon substrate with a 90-nm-thick SiO₂ layer that worked as back-gate electrode/dielectric. We measured the thickness of exfoliated ReSe₂ bulk flakes through atomic force microscopy (AFM). It was approximately 80 nm (around 115 layers), and ReSe₂ flakes with similar thickness were selected for this experiment. These samples were washed in acetone and deionized water to remove the tape residues. An optical lithography process was then performed to pattern the source/drain electrodes on the selected flakes, where the channel width and length were 5 μm each. Ti (10 nm) and Pd (50 nm) layers were deposited using an e-beam evaporator, followed by a lift-off process. Due to device-to-device variation, we fabricated many devices and then selected six devices with very similar on-current and threshold voltages (three device samples for 1% and 5% conditions, respectively).

2.2. APTMS doping of ReSe₂

For the formation of the APTMS layer, toluene (50 mL), and APTMS solutions (500 μL equivalent 1 wt%, and 2500 μL equivalent 5 wt%) were prepared, and the ReSe₂ devices were immersed in each solution for 40 min to form an APTMS self-assembled monolayer (SAM) on top of the ReSe₂ channel region. After rinsing in toluene and deionized water several times, the samples were baked in ambient air for 20 min at 120 $^{\circ}\text{C}$.

2.3. Characterization of ReSe₂ devices

Raman analysis was performed on ReSe₂ thin films using alpha300 M||WITec spectroscopy to characterize the effects of APTMS doping. Raman spectroscopy utilized a 532-nm excitation wavelength with a 0.7- to 0.9- μm beam size and instrumental spectral resolution less than 0.9 cm^{-1} . A spectrometer having 1800 grooves/mm was applied with a 5-s integration time. Furthermore, electrical measurements (I_D - V_G and I_D - V_D) were carried out using an HP 4155A semiconductor parameter analyzer. In particular, the I_D - V_G measurement was conducted under both dark and light illumination conditions before and after the APTMS doping. We used dot lasers with different wavelengths (520 nm, 655 nm, 785 nm, 850 nm, 980 nm, and 1064 nm) and optical power of 1 mW cm^{-2} . Moreover, the power was adjusted to 27 pW, 70 pW, 265 pW, 850 pW, 2020 pW, and 15300 pW using a “mounted round variable ND filter.” The carrier concentration (n) was extracted using the equation $n = I_D L / q W \mu V_D$, where L and W are the length and width of the ReSe₂ channel, respectively, q is the electron charge, and C_{OX} is the gate capacitance per unit area. Additionally, the extracted carrier concentration was normalized by dividing by the number of ReSe₂ layers, and the result was expressed in unit of cm^{-2} . The photoresponsivity (R) values were extracted utilizing the equation $R = I_{\text{Photo}} / P_{\text{Light}}$, where I_{Photo} and P_{Light} represent the generated photocurrent and incident optical laser power, respectively. All the measurements were performed in air.

3. Results and discussion

First, Raman spectroscopy analysis was performed to investigate the effects of APTMS treatment on the ReSe₂ films. Fig. 1a shows the Raman spectra for ReSe₂ before and after the 5 wt% APTMS treatment. Many Raman peaks between 160 cm^{-1} and 270 cm^{-1} were observed on the ReSe₂ samples, unlike other conventional TMDs with isotropic hexagonal structure such as MoS₂ and WSe₂. This is because ReSe₂ has 36 phonon modes (including 18 Raman-active vibrational modes) derived from a distorted triclinic structure consisting of 12 nonequivalent

atoms [25]. These Raman peaks were slightly red-shifted after treating ReSe₂ with APTMS, as represented by the dotted gray and red lines in Fig. 1a. This is apparently attributed to the n-type doping effect, based on the previous report that n-type doping of ReSe₂ using triphenylphosphine (PPh₃) triggered red-shift for an A_g-like peak by -1.29 cm^{-1} [26]. The schematic explaining the n-doping of ReSe₂ by APTMS is presented in Fig. 1b. When a ReSe₂ film is treated by APTMS, the negative charges from amine functional groups ($-\text{NH}_2$) in APTMS attract holes on the APTMS/ReSe₂ interface, relatively increasing the electron carrier density in the ReSe₂ film [27]. In addition, to quantitatively investigate the Raman peak shifts according to the concentrations of APTMS solutions (1 and 5 wt%), we extracted the red-shift values for the two high intensive peaks at 174 cm^{-1} and 263 cm^{-1} , as shown in Fig. 1c. The red-shift values for the peaks increased from -0.68 cm^{-1} to -1.41 cm^{-1} and from -0.86 cm^{-1} to -1.85 cm^{-1} respectively, as the APTMS concentration increased from 1 wt% to 5 wt %, demonstrating that the doping level is controlled by the adjustment of APTMS concentration in solution. Fig. 1d shows the Raman mapping images corresponding to the high intensive peaks of ReSe₂, before and after the 5 wt% APTMS treatment. The darker ReSe₂ images were obtained for both cases of 174 cm^{-1} and 263 cm^{-1} peaks after the APTMS treatment, which also supports that APTMS induced the n-type doping phenomenon over the entire area of ReSe₂. Here, low root mean square (RMS) values (0.19–0.21 cm^{-1} for pristine and 0.21–0.25 cm^{-1} for the APTMS-doped ReSe₂ layers) were observed, meaning that ReSe₂ was uniformly n-doped (Fig. 1e).

Next, we fabricated ReSe₂-based transistors (Fig. 2a) and performed electrical measurement to investigate the influence of APTMS n-doping on the electrical characteristics (on-current, V_{TH} , and carrier concentration). Fig. 2b shows the I_D - V_G transfer characteristics (at $V_{\text{DS}} = 5 \text{ V}$) of undoped (solid line) and 5 wt% APTMS-doped (dotted line) ReSe₂ devices with channel length and width of 5 μm . After the APTMS treatment (5 wt%), the on-current increased three-fold (from 1.20×10^{-7} to $3.60 \times 10^{-7} \text{ A}/\mu\text{m}$ at $V_{\text{GS}} = 30 \text{ V}$) and a negative shift occurred in the threshold voltage (from -3 V to -6.8 V). These changes seem to originate from the decrease in the effective barrier height (Φ_{eff}) at the Ti-ReSe₂ junction by the n-type doping. When the APTMS was coated on the ReSe₂ channel, the negative charges of APTMS amine functional groups attracted holes at the interface between APTMS and ReSe₂, thereby increasing the electron carrier concentration in the ReSe₂ and shifting down the energy band of ReSe₂. This phenomenon reduces the effective barrier height at the Ti-ReSe₂ junction and raises the probability of electron carrier injection (as presented in Fig. 2c), consequently increasing the on-current and shifting the threshold voltage negatively. Particularly, this barrier lowering by n-doping can be intensified by increasing the concentration of APTMS solution ($\Phi_{\text{eff},1\%} > \Phi_{\text{eff},5\%}$). As shown in Fig. 2d and e, the I_{ON} ratio and ΔV_{TH} increased from 1.31 to 2.41 A/A (1 wt%: $3.60 \times 10^{-8} \rightarrow 4.27 \times 10^{-8} \text{ A}/\mu\text{m}$ and 5 wt%: $1.20 \times 10^{-7} \rightarrow 3.60 \times 10^{-7} \text{ A}/\mu\text{m}$) and from -1.7 V to -4.4 V , respectively, as the APTMS concentration was changed from 1 to 5 wt%. Moreover, the extracted doping concentrations (Δn) of ReSe₂ samples doped by 1 and 5 wt% APTMS solutions were 7.82×10^{10} and $3.90 \times 10^{11} \text{ cm}^{-2}$, respectively. These values are comparable to the concentrations achieved by the previous TMD doping techniques, such as Cs₂CO₃ n-doping of MoS₂ by Lin et al. (approx. 0.5 – $3.5 \times 10^{11} \text{ cm}^{-2}$) [28], thiol-based FODT molecular p-doping of MoS₂ by Sim et al. (approx. $7 \times 10^{11} \text{ cm}^{-2}$) [29], and OTS p-doping of WSe₂ by Kang et al. (approx. 2.1 – $5.2 \times 10^{11} \text{ cm}^{-2}$) [19]. This n-doping phenomenon by APTMS likewise lies in the nondegenerate regime, thereby maintaining the semiconductor properties of ReSe₂, whereas TMDs behaves as near-metallic layers in high concentration doping (degenerate) [30].

We then investigated I_D - V_G characteristics of ReSe₂ photodetectors under dark and laser-illuminated conditions. The structure of the ReSe₂ photodetector is depicted in Fig. 3a, along with the energy band diagram for the source-TMD junction (Ti-ReSe₂) under light illumination,

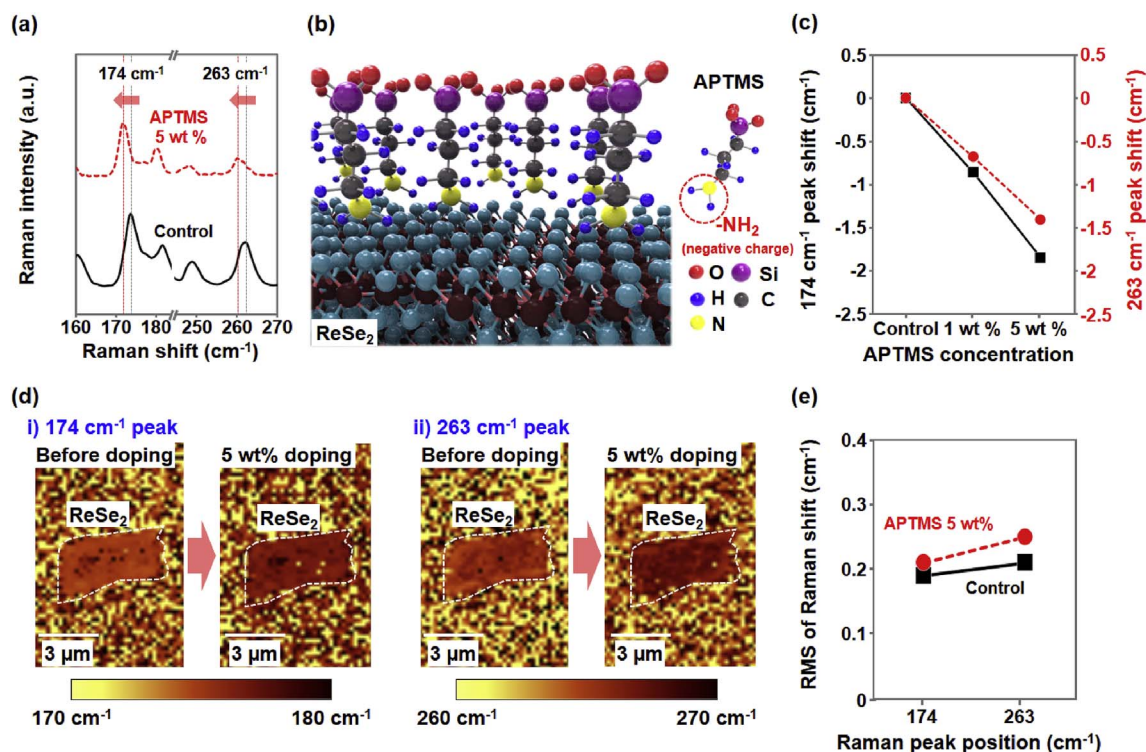


Fig. 1. (a) Raman spectra of pristine (solid line) and APTMS-doped (dotted line) ReSe₂. (b) Graphical description of the mechanism of APTMS n-doping on the ReSe₂ film. (c) Extracted Raman peak shift values (at 174 and 263 cm⁻¹) for 1 and 5 wt% concentrations. (d) Raman mapping images and (e) Root mean square (RMS) values of Raman peak shift (at 174 and 263 cm⁻¹) before and after doping with APTMS 5 wt% concentration.

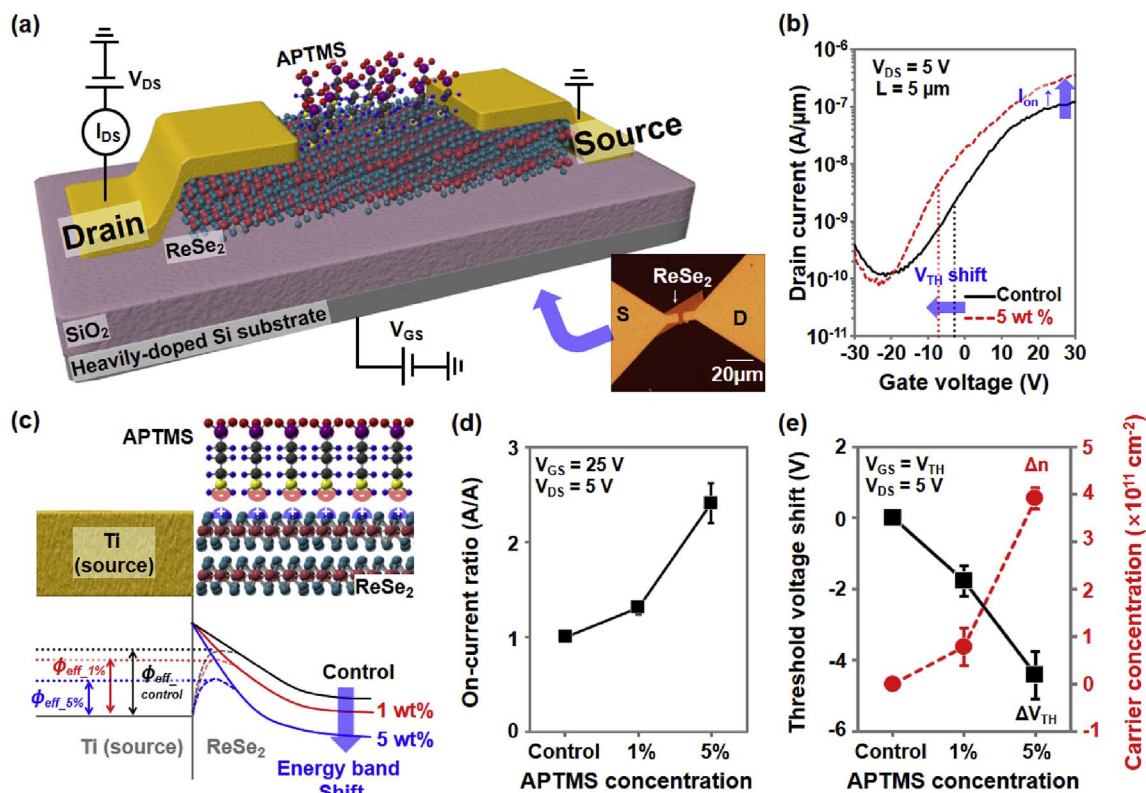


Fig. 2. (a) Schematic diagram of APTMS-doped ReSe₂ photodetector and optical image of the device. (b) I_D - V_G curves for control (solid line) and APTMS-doped (dotted line) ReSe₂ photodetectors ($V_{DS} = 5$ V). (c) Energy band diagram of Ti-ReSe₂ junction before and after APTMS doping (1 and 5 wt%). (d) On-current ratio, (e) threshold voltage shift (solid line) and carrier concentration (dotted line) before and after doping with 1 and 5 wt% APTMS concentrations. Here, on-current and carrier concentration were extracted at $V_{GS} = 25$ V and $V_{GS} = V_{TH}$, respectively.

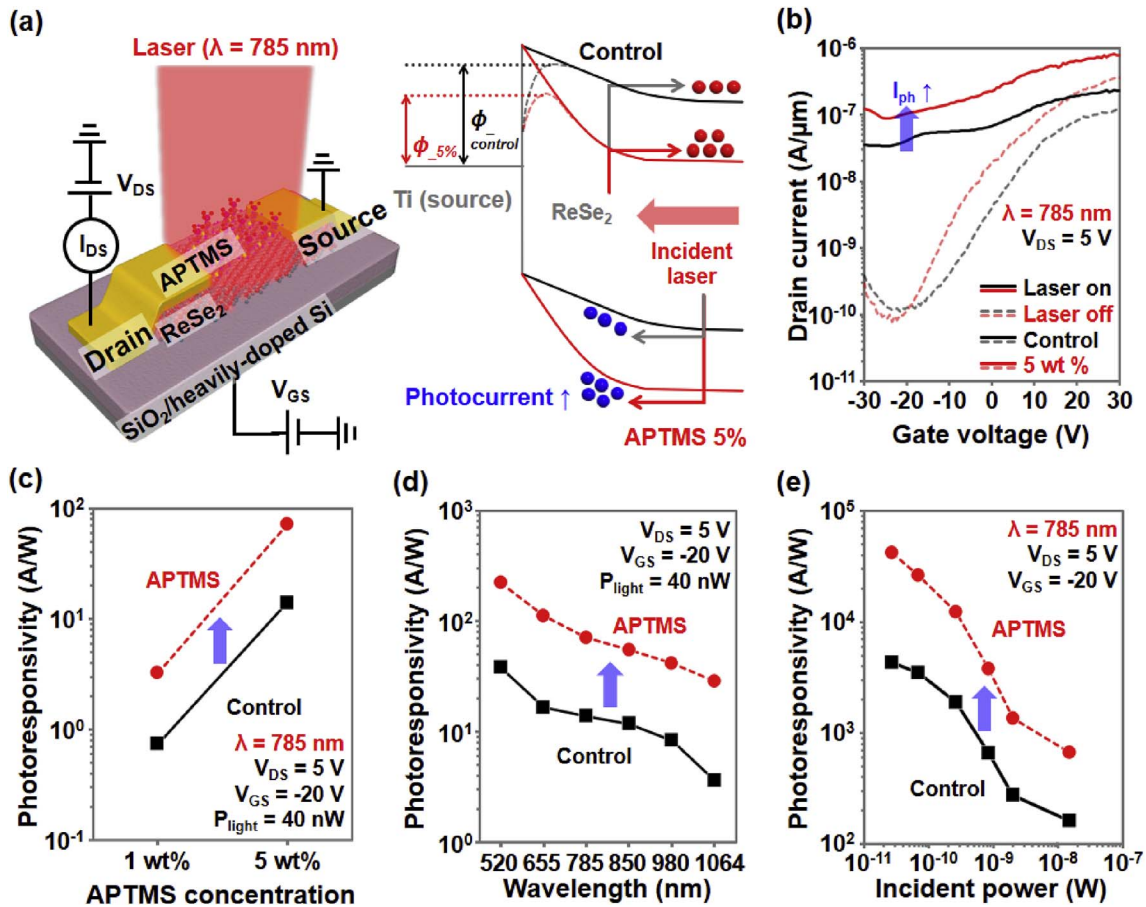


Fig. 3. (a) Schematic diagram of the APTMS-doped ReSe₂ photodetector and energy band diagram at source-ReSe₂ junction under laser exposure. (b) I_D - V_G curves before and after laser exposure for undoped and APTMS-doped ReSe₂ photodetector. Extracted photoresponsivity values before and after doping as a function of (c) APTMS concentration, (d) laser wavelength, and (e) incident laser power. Here, the photoresponsivity values were extracted at $V_{GS} = -20$ V.

where the bandgap of ReSe₂ is 1.0 eV. When the n-doping is induced on the ReSe₂ photodetector by APTMS, the effective electron barrier height at the Ti-ReSe₂ junction is lowered, and the contact resistance decreases, consequently increasing the electric field near the junction. This enhancement in electric field probably increased the number of photocarriers collected in the ReSe₂ channel, thus improving the photocurrent ($I_{\text{photo}} = I_{\text{laser, on}} - I_{\text{laser, off}}$) from 5.9×10^{-8} A/ μm to 1.57×10^{-7} A/ μm at $V_{GS} = V_{\text{TH}}$ (Fig. 3b). For further analysis, the photoresponsivity ($R = I_{\text{photo}}/P_{\text{light}}$) was extracted from the I_D - V_G curves measured under different doping and laser conditions (APTMS concentration, light wavelength, and light incident power). Fig. 3c shows the photoresponsivity values of the ReSe₂ photodetector before/after APTMS doping ($\lambda = 785$ nm, $P_{\text{light}} = 40$ nW/cm²). When 1 and 5 wt% APTMS doping was applied, the photoresponsivity was remarkably improved (1 wt%: 0.74 $\rightarrow$ 3.2 A/W, 5 wt%: 14 $\rightarrow$ 71.7 A/W), where the higher APTMS concentration showed a profound effect on the photoresponsivity because the stronger n-doping effect further enhanced the photocarrier collection by reducing the contact resistance and increasing the electric field at the Ti-ReSe₂ junction. The photoresponsivity was improved across all laser wavelengths, whereby the highest photoresponsivity of 221 A/W was obtained at 520-nm laser wavelength after 5 wt% APTMS doping, which is considerably higher than the previously reported photoresponsivity values for ReSe₂ photodetectors [15,16]. In particular, the photoresponsivity obtained at the near-infrared laser wavelength of 1064 nm increased eightfold from 3.6 A/W to 28.5 A/W after the APTMS treatment, which is much higher than the values of recently reported black phosphorous- [12] and molybdenum ditelluride (MoTe₂)-based [31] infrared photodetectors (5 mA/W at 1550 nm and 24 mA/W at 1060 nm, respectively). In

addition, the cutoff wavelength confirmed in this experiment ($\lambda_{\text{cutoff, ReSe}_2}$: around 1064 nm) seems to be higher than λ_{cutoff} reported in previous studies for MoS₂ [10] and WSe₂ [11], whereby λ_{cutoff} : around 980 nm. This demonstrates the broad detection capability of ReSe₂ from visible to near-infrared regions due to its lower bandgap. The effect of incident light power (27 pW, 70 pW, 265 pW, 850 pW, 2020 pW, and 15300 pW) on the photoresponsivity is also illustrated in Fig. 3e. It was found that the photoresponsivity increases as the light power decreases in both control and APTMS-doped ReSe₂ photodetectors. The highest photoresponsivity was measured at 27 pW laser power (wavelength: 785 nm), where the photoresponsivity also increased tenfold from 4.3×10^3 A/W to 4.2×10^4 A/W after APTMS 5 wt% n-doping. One plausible reason for this disparity could be the suppressed scattering phenomenon due to the fewer photogenerated carriers under lower light power [32].

Finally, to investigate the APTMS n-doping effect on the ReSe₂ photodetector in terms of the photoswitching time (rising and decaying times), we performed temporal light response measurement on the ReSe₂ photodetector by chopping the 785-nm laser with power = 40 nW/cm² in consecutive 20-s intervals (Fig. 4a). Fig. 4b shows the photocurrents as a function of laser exposure time, which were obtained in ReSe₂ photodetectors before and after APTMS 5 wt% n-doping (at $V_{DS} = 5$ V and $V_{GS} = 0$ V). Here, the photocurrent was remarkably improved by applying APTMS n-doping (from approximately 0.13 to 0.35 $\mu\text{A}/\mu\text{m}$) while maintaining the fast photoswitching time. The temporal photoresponse curves magnified at the rising and decaying edges are shown in Fig. 4c and d. The photoresponse rising and decaying times were obtained by finding difference in the time points at 10% and 90% of the measured maximum photocurrents. After

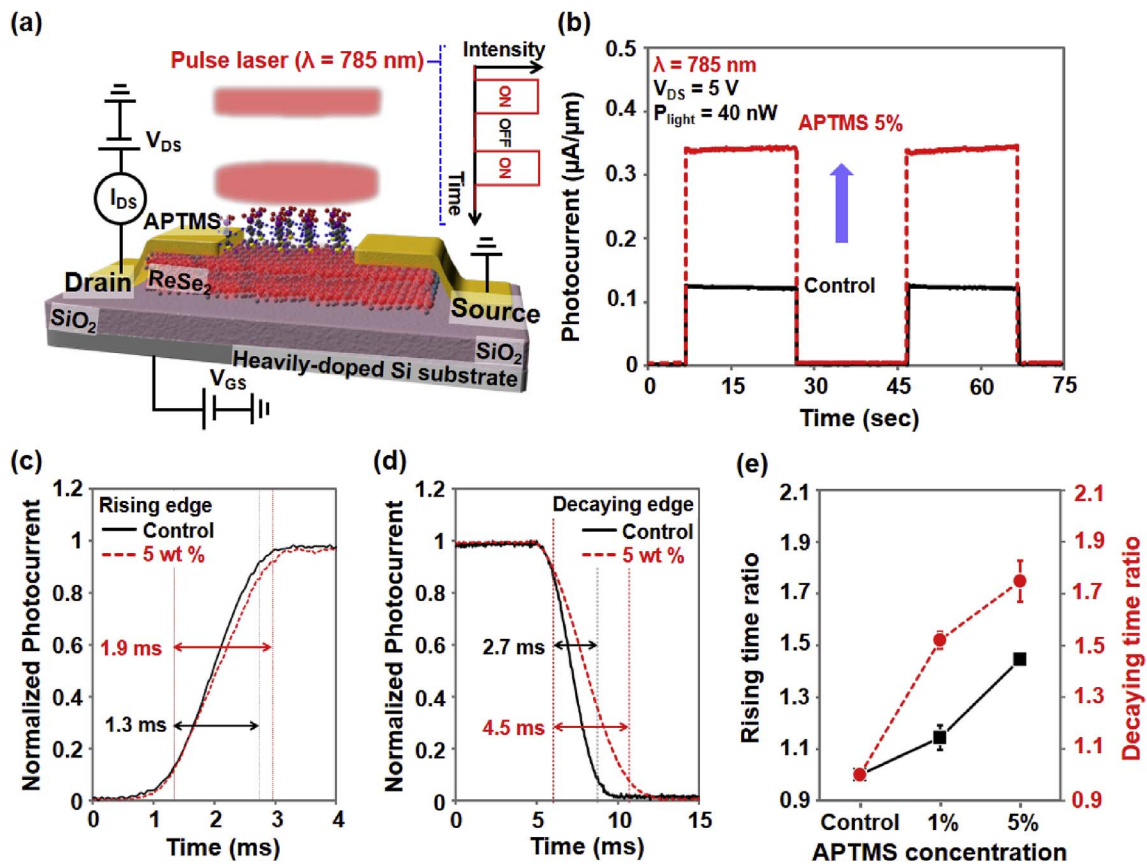


Fig. 4. (a) Schematic diagram for photoresponse measurement of the undoped and APTMS-doped ReSe₂ photodetector (b) Temporal light response characteristic for undoped (solid line) and APTMS-doped (dotted line) ReSe₂ photodetectors. The temporal photoresponse curves at the (c) rising and (d) decaying edges. (e) The extracted rising time ratio (solid line) and decaying time ratio (dotted line) for 1 and 5 wt% APTMS concentrations.

the 5 wt% APTMS treatment, both the rising and decaying times slightly increased; $\tau_{\text{rise}}/\tau_{\text{decay}}$ were 1.3/2.7 ms for control and 1.9/4.5 ms for 5 wt% APTMS. This increase in the photoswitching times after the APTMS doping was also dependent on the concentration of APTMS in solution. As the APTMS concentration changed from 1 to 5 wt %, the rising time ratio ($\tau_{\text{rise_APTMS}}/\tau_{\text{rise_Control}}$) and the decaying time ratio ($\tau_{\text{decay_APTMS}}/\tau_{\text{decay_Control}}$) increased from 1.17 to 1.44 (1 wt %: 1.03 $\rightarrow$ 1.2 ms and 5 wt%: 1.3 $\rightarrow$ 1.9 ms) and from 1.53 to 1.74 (1 wt %: 3.3 $\rightarrow$ 5.06 ms and 5 wt%: 2.7 $\rightarrow$ 4.5 ms), respectively (Fig. 4e). Here, the error bars were obtained by measuring three different device samples for each doping condition (1 and 5 wt %). This can be explained by the shrinkage in depletion width at the Ti-ReSe₂ interface due to the APTMS doping. When APTMS n-doping is applied to ReSe₂ photodetectors, the depletion width is expected to be narrowed because the effective barrier height at the Ti-ReSe₂ junction is lowered by the n-doping, consequently making it difficult to separate the photogenerated carriers in the ReSe₂ region [26]. Although the APTMS n-doping effect slightly decreased the photoswitching time of ReSe₂ photodetectors, the rising/decaying times of our APTMS-doped ReSe₂ photodetector ($\tau_{\text{rise}}/\tau_{\text{decay}}$: 1.9/4.5 ms) were approximately ten times shorter than previously reported values for rhenium-based TMDs ($\tau_{\text{rise}}/\tau_{\text{decay}}$: 68/34 ms for ReSe₂ [15] and 0.67/5.6 s for ReS₂⁹).

4. Conclusion

In conclusion, we have demonstrated a high-performance ReSe₂ photodetector with high photoresponsivity, wide detection range, and short photoswitching time using an APTMS n-doping technique. The APTMS n-doping effect originated from the amine functional groups ($-\text{NH}_2$) in APTMS, which have negative charges attracting holes at the APTMS/ReSe₂ interface and relatively increase the electron carrier

density in the ReSe₂. The n-doping concentration of ReSe₂ was adjusted between 7.82×10^{10} and $3.90 \times 10^{11} \text{ cm}^{-2}$, which lies in the non-degenerate regime. We investigated the n-doping effect by APTMS treatment via Raman spectroscopy and electrical and optical measurements. Because of the reduced effective barrier height at the Ti-ReSe₂ junction by the APTMS n-doping, a negative shift was induced in the threshold voltage (from -3 to -6.8 V), and the on-current ratio was improved approximately threefold (from 1.20×10^{-7} to $3.60 \times 10^{-7} \text{ A}/\mu\text{m}$). In addition, the photoresponsivity values were dramatically improved at all wavelengths between 520 and 1064 nm after performing the APTMS doping technique (approximately tenfold at 785 nm, from 4.3×10^3 to $4.2 \times 10^4 \text{ A}/\text{W}$ and eightfold at 1064 nm, from 3.6 to 28.5 A/W). The photoswitching times of APTMS-doped ReSe₂ photodetectors ($\tau_{\text{rise}}/\tau_{\text{decay}}$: 1.9/4.5 ms) were approximately ten times shorter than previously reported values in the devices formed on rhenium-based TMDs. We believe that our doping technique could be utilized to effectively improve the performance of various 2D optoelectronic applications in the future.

Acknowledgement

This research was supported by the Basic Science Research Program, Basic Research Lab Program, and Nano-Material Technology Development Program through National Research Foundation of Korea (NRF) grants funded by the Korea government (MSIP) (NRF 2015R1A2A2A01002965, 2015M3A7B7045496, 2017R1A4A1015400 and 2016M3A7B4910426), and the Future Semiconductor Device Technology Development Program (10067739) funded by Ministry of Trade, Industry and Energy (MOTIE), and Korea Semiconductor Research Consortium (KSRC).

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